

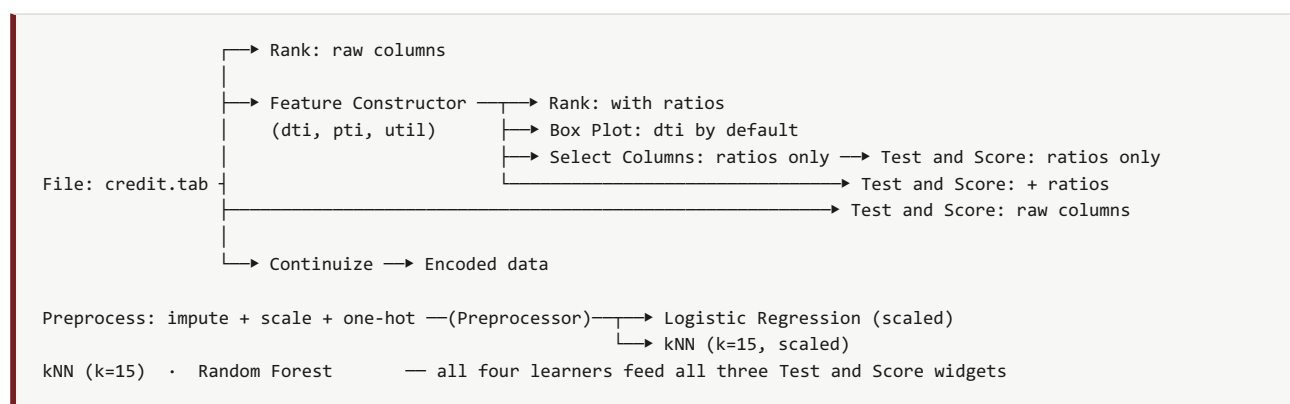
Week 5 — week05_features_pca.ows

Companion notes for the Orange workflow. Book: chapter 5. Notebook twin: `notebooks/week05_features.ipynb`.

1. What this workflow is

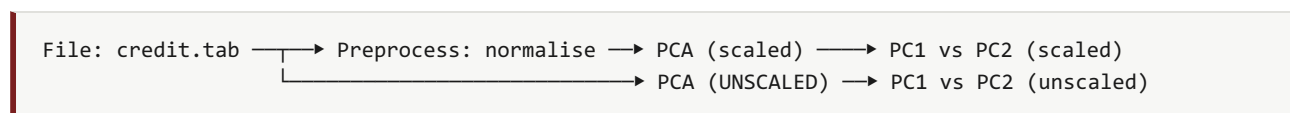
Two **controlled experiments on the canvas**, one above the other. In each, the branches receive the same data and differ in exactly one thing, so every difference downstream is caused by that one thing.

Part A — try things with features (top of the canvas).



Same four learners, same 5 folds, three different sets of columns. The *only* thing that changes between the three Test and Score widgets is the features.

Part B — PCA needs scaling (bottom of the canvas).



Dataset — `data/credit.tab`, the same 4,000 loan applications, with the money columns in raw rupiah. That matters in both parts: `loan_amount` has a standard deviation around 29 million, `age` around 13. Any method built on distances or variances cannot ignore a factor of two million.

The ideas being taught.

- *Part A*: a well-chosen feature can be worth more than a better model — but whether a model can *use* it depends on the model. Distance-based models also need the feature on a comparable scale, and are hurt by irrelevant columns.
- *Part B*: PCA maximises *variance*, and variance is not scale-free. Feed it raw rupiah and it reports the column with the biggest units.

Watch the channel names. Each is a hint about what the widget produces:

- Preprocess emits `Preprocessor` (a recipe, sent *into* a learner) and `Preprocessed Data` (a transformed table). Part A uses the first, Part B the second — the same widget, used two different ways.
- Rank emits `Reduced Data`, not `Data`.
- PCA emits `Transformed Data` (the projected coordinates) as well as `Components` and `Data`.

Pre-configured, and why.

- *Feature Constructor* already holds the three ratios, with guarded denominators:
`debt / max(income, 1), monthly_payment / max(income / 12, 1), balance / max(credit_limit, 1)`.
- *Select Columns* keeps only `dti`, `pti`, `util` as features.
- *Box Plot* shows `dti` split by `default`.
- *kNN* uses $k = 15$ (as in the notebook); *Random Forest* uses 100 trees and a fixed seed, so everyone sees the same numbers.
- *Preprocess: impute + scale + one-hot* feeds two learners. **A preprocessor sent to a learner replaces that learner's own defaults** (imputation and one-hot encoding) rather than adding to them. Send only "normalise" and logistic regression fails on the first categorical column or missing income. So the chain restates the defaults around the scaling. Passing the preprocessor *into* the learner is also the leak-free wiring: it is refitted on each training fold, exactly like the notebook's `Pipeline`.
- *Part B*: the Preprocess widget has *Normalize Features* enabled (mean 0, sd 1), and **both PCA widgets have their own *Normalize variables* checkbox switched off**. Orange's PCA widget normalises by default; left on, it would quietly scale the "UNSCALED" branch too and both branches would report identical variance. With it off, Preprocess is the only difference. Toggle Preprocess off and watch the contrast collapse — or tick *Normalize variables* in the UNSCALED PCA — to prove the effect to yourself.

2. What to do

Part A — features

1. **Compare the two Rank widgets.** Where do `dti`, `pti` and `util` land once they exist? Where were `debt` and `income` before?
2. **Box Plot.** Does `dti` look different for defaulters? Switch the variable to `debt`, then `income`.
3. **Open the three Test and Score widgets**, sort by AUC, and fill in a 4×3 table. Which model gains most from the ratios? Which gains least?
4. **Unscaled kNN** scores exactly the same with and without the ratios, while scaled kNN gains the most. Why can unscaled kNN not see them?
5. **Ratios only** — 3 features instead of 16. Unscaled kNN suddenly jumps. What did removing the rupiah columns do to its distances?

6. **Try your own.** Add a fourth expression in Feature Constructor, e.g.

`loan_amount / max(income, 1)`, and watch all three Test and Score widgets update.

7. **Continue.** Switch the treatment of categorical columns (one-hot, ordinal, remove) and look at the *Encoded data* table.

Part B — PCA

1. **Preprocess** — confirm *Normalize Features* is on.

2. **Open both PCA widgets.** How many components are needed to reach 95% of the variance in each?

3. **Compare the two scatter plots.** One of them is essentially a plot of a single variable. Which, and why? See book §5.6.4, limitation 2.

Versi mahasiswa. Pembahasan, hal yang perlu diperhatikan, dan jawaban setiap langkah disampaikan di kelas setelah praktikum.